



THE MATHEMATICS AND PHYSICS OF ODM GEAR

A Physical Model of Levi Ackerman's Three-Dimensional Movement

From Grappling Cables to Constrained Dynamics, Energy, and Trajectory Optimisation

Abstract

Levi Ackerman, Captain of the Survey Corps in Attack on Titan, moves through three-dimensional urban environments with a precision and speed that appears to defy ordinary mechanical intuition. His Omni-Directional Mobility (ODM) gear — a system of compressed-gas propulsion and dual steel-wire anchors — is the physical mechanism behind every aerial combat manoeuvre the series depicts. This paper asks a serious mechanical question: can Levi's characteristic three-dimensional ODM movement be represented as a physically meaningful constrained dynamical system, and what do mathematics and Newtonian mechanics predict about speed, acceleration, cable tension, energy, trajectory, and physical survivability? The central mathematical object is the cable-length constraint $\|\mathbf{r}_L(t) - \mathbf{r}_A(t)\| = L(t)$, which encodes the complete geometry of tethered motion. Starting from elementary kinematics and progressing through vector mechanics, centripetal dynamics, dual-anchor control theory, aerodynamic drag, rotational mechanics, material stress, human acceleration physiology, and constrained trajectory optimisation, the paper traces a mathematical narrative in which each new framework becomes necessary precisely because the previous one cannot fully account for what is observed. The analysis distinguishes rigorously between what real physics can explain and where fictional technology must be assumed.

RESEARCH QUESTION

Can Levi Ackerman's three-dimensional ODM movement be represented as a physically meaningful constrained dynamical system, and what do mathematics and mechanics predict about the speed, acceleration, cable tension, energy budget, trajectory geometry, and human survivability of each manoeuvre? Where does the model succeed, and where does the fictional system exceed the limits of real-world physics?

1 Introduction

In Episode 22 of *Attack on Titan*, Levi Ackerman faces the Female Titan — a creature approximately 14 metres tall, armoured, and regenerating. His response is not to retreat. He deploys both ODM anchors, arcs through the air in a tight curve around the Titan's head, changes direction in mid-flight, and lands three successive strikes in under two seconds. Watching the sequence, the physical question is immediate and unavoidable: what force bent his trajectory? What tension pulled the cables? What energy powered the manoeuvre? And, most fundamentally — could a human body survive the accelerations involved?

These are not rhetorical questions. They are mechanical questions, and they have mechanical answers — or, at least, they have the kind of principled estimates and constraint analyses that serious engineering produces when exact data are unavailable. This paper attempts those answers. The ODM gear is treated as a real physical system: a dual-cable tethered platform with gas propulsion, operating under Newtonian mechanics. The analysis does not pretend to have access to cable diameters, gas pressures, or anchor penetration forces that the series never specifies. Where canon provides data, it is used. Where it does not, engineering assumptions are introduced explicitly, labelled, and tested for sensitivity.

The key mathematical idea that organises the entire paper is a single constraint equation:

$$\|\mathbf{r}_L(t) - \mathbf{r}_A(t)\| = L(t)$$

This expression — that Levi's distance from the anchor point equals the cable length — is the bridge between the anime and serious mechanics. From it, the paper derives velocity constraints, acceleration components, cable tension requirements, energy expenditure, and ultimately the conditions under which the system would fail. The mathematics escalates progressively: each new framework becomes necessary because the previous one is unable to fully account for the observed phenomenon. This structure is the same intellectual architecture that makes theoretical physics compelling — not a catalogue of facts, but a sequence of problems and solutions.

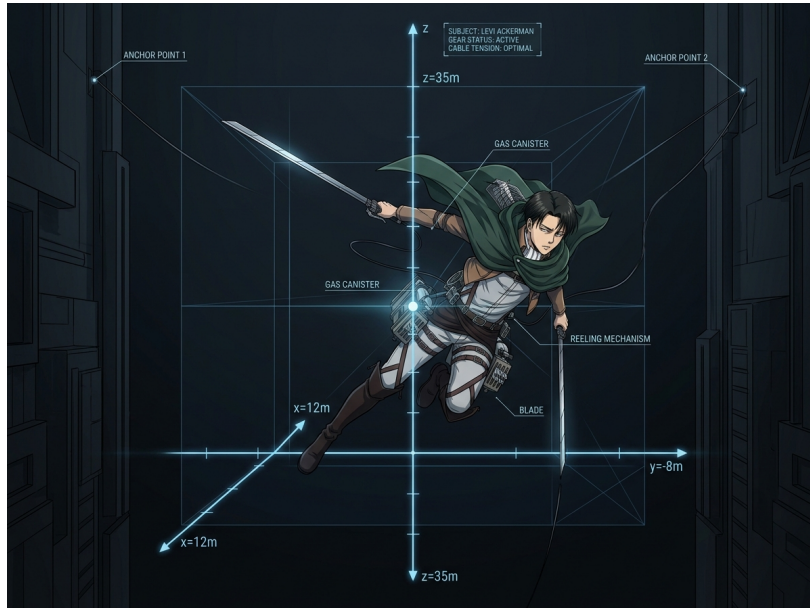


Figure 1: Levi Ackerman positioned in a 3D coordinate system. ODM combat is inherently three-dimensional — buildings provide anchor points at arbitrary spatial locations, and Levi's position vector $\mathbf{r}_L(t) = [x(t), y(t), z(t)]$ sweeps a curved path through this space. No one-dimensional model can capture the geometry.

2 What Is ODM Gear?

Canonical mechanism and component analysis

The Omni-Directional Mobility gear — also described in the series as "Three-Dimensional Manoeuvre Equipment" — is the principal combat technology of the Survey Corps. Its function is to enable soldiers to move through three-dimensional environments at high speed, primarily in urban settings where buildings provide the elevated anchor surfaces that the system requires.

Component	Canon Description	Physical Role
Body Harness	Leather and metal strap system; distributes load to torso, hips, thighs	Transmits cable tension to skeletal frame; reduces point-load injury risk
Gas Canisters	Pressurised Iceburst Stone gas; hip/lower-back mounted	Powers turbine reel; provides auxiliary thrust; energy reservoir
Wire Reel Housings	Side-mounted on waist; contain steel wire coils; fires/retracts anchors	Issues constraint $L(t)$; controls swing radius; converts gas energy to kinetic energy
Anchor Hooks	Steel grappling anchors; fired into structures or Titan flesh	Provides attachment point r_A ; transfers tension force to structure
Control Grips / Hilt	Integrated into sword hilts; trigger-operated	User interface: controls anchor fire, retraction rate, gas release
Blades	"Ultrahard steel"; thin, replaceable	Cutting instrument; mass $\approx 0.3\text{--}0.5\text{ kg}$ [ASSUMPTION]

Table 1: ODM Gear Component Summary. [CANON] entries derived from official Attack on Titan series descriptions and guidebooks. [ASSUMPTION] entries are modelling inputs introduced for calculation purposes and explicitly labelled.

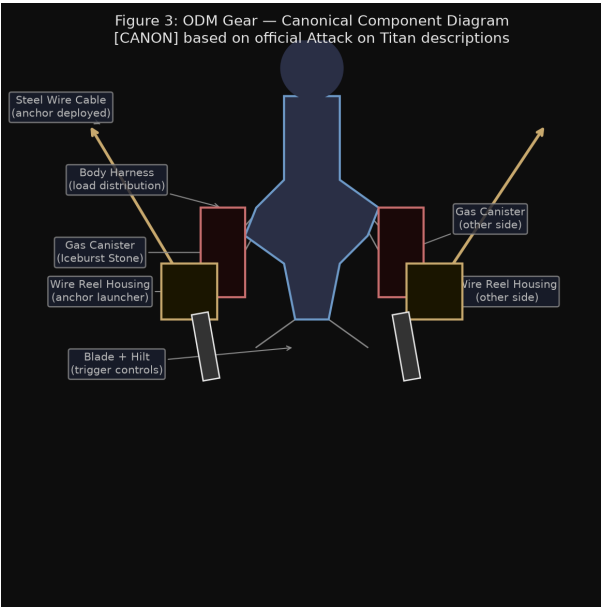


Figure 2: Schematic diagram of ODM gear components. The system integrates gas propulsion, wire-reel mechanics, anchor deployment, and blade-hilt controls into a single wearable platform. [CANON] labels mark features established by the series; [MODEL] labels indicate mechanical interpretations introduced in this paper.

Two points are critical for the mechanical analysis that follows. First, the gas provides both reel-driving power and directional thrust. These are functionally separate: reel power changes cable length $L(t)$, while thrust adds a propulsive force vector to the equations of motion. Second, the dual-cable architecture is not decorative — it is mechanically necessary for three-dimensional steering. A single-cable system can only swing in a plane; two independently controlled cables allow lateral redirection, rotation, and braking. Section 6 analyses this in detail.

3 From Anime Motion to Kinematics

Position, velocity, acceleration, and the vector description

Before the cable constraint can be introduced, the most elementary kinematic description must be established. Motion in space requires a coordinate system. Assign to Levi's centre of mass a position vector:

$$\mathbf{r}(t) = [x(t), y(t), z(t)]$$

where x and y are horizontal coordinates and z is height above a chosen reference plane. Velocity is the rate of change of position:

$$\mathbf{v}(t) = d\mathbf{r}/dt = [\dot{x}(t), \dot{y}(t), \dot{z}(t)]$$

and acceleration is the rate of change of velocity:

$$\mathbf{a}(t) = d\mathbf{v}/dt = d^2\mathbf{r}/dt^2 = [\ddot{x}(t), \ddot{y}(t), \ddot{z}(t)]$$

The magnitude of velocity — the speed — is:

$$v(t) = |\mathbf{v}(t)| = \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}$$

For an anchor point A at fixed position $\mathbf{r}_A = [x_A, y_A, z_A]$, the cable vector pointing from Levi toward the anchor is:

$$\mathbf{c}(t) = \mathbf{r}_A - \mathbf{r}_L(t)$$

and the cable length is:

$$L(t) = |\mathbf{c}(t)| = |\mathbf{r}_A - \mathbf{r}_L(t)|$$

This is not yet a constraint. It is simply a geometric fact. The constraint arises when the reel locks: if no wire is being paid out or retracted, $L(t)$ must remain constant, and Levi's motion is restricted to a sphere of radius L centred on the anchor. This is the simplest model of ODM motion, and the next section examines it carefully before discarding it in favour of a richer framework.

A note on Levi's physical parameters. The official *Attack on Titan* guidebooks record Levi's height as 160 cm and mass as 65 kg [CANON]. The ODM gear adds mechanical mass. No official mass is given for the gear, but engineering reasoning suggests a range of 10–15 kg [ASSUMPTION: comparable to a loaded military pack plus mechanical hardware]. Throughout this paper, the combined mass is taken as $m = 80$ kg as a representative central estimate, with sensitivity noted where the result changes significantly.

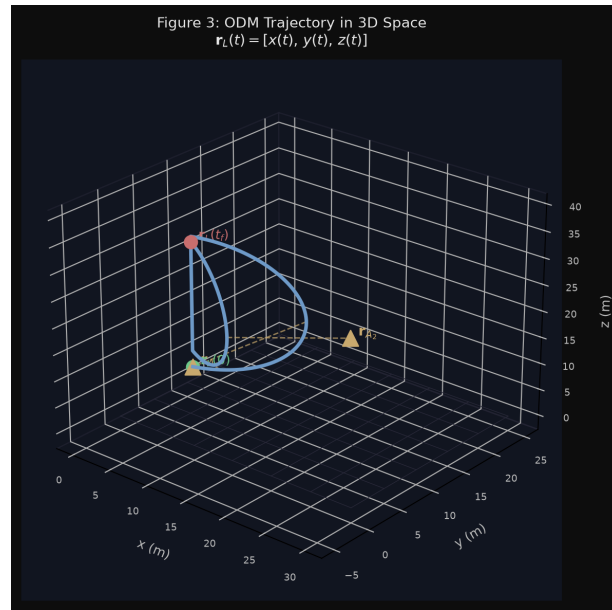


Figure 3: A representative two-anchor ODM trajectory in three-dimensional space. The path (blue) is not a straight line, not a simple arc, and not confined to a single plane. Anchor positions $\mathbf{r}_{A1}$ and $\mathbf{r}_{A2}$ (triangles) are fixed; Levi's position $\mathbf{r}_L(t)$ (circle: start green, end red) sweeps a compound curve. Modelling this path correctly requires the full three-dimensional vector description of Section 3.

4 The Cable as a Mathematical Constraint

Fixed and variable tether length

The cable introduces a geometric constraint on Levi's position. This is conceptually different from a force. A constraint is a restriction on which positions and velocities are physically allowed, given the mechanical configuration. For a cable of length $L(t)$ attached to anchor A, the constraint is simply:

$$\|\mathbf{r}_L(t) - \mathbf{r}_A\| = L(t)$$

■ CENTRAL CONSTRAINT

This single equation is the central mathematical object of the paper. It encodes Levi's distance from the anchor point, the cable geometry, and the mechanical coupling between his trajectory and the anchor location. Every result in the sections that follow — velocity, acceleration, tension, energy — can be traced back to this constraint.

Differentiating the constraint with respect to time (treating L as possibly time-varying) gives a velocity constraint. Let $\mathbf{u} = (\mathbf{r}_A - \mathbf{r}_L) / L$ be the unit vector along the cable. Then differentiating $\|\mathbf{r}_L - \mathbf{r}_A\|^2 = L^2$ gives:

$$(\mathbf{r}_L - \mathbf{r}_A) \cdot \mathbf{v}_L = L \cdot dL/dt$$

This tells us that the component of Levi's velocity along the cable direction equals the rate at which cable is being reeled out or in. If the cable length is fixed ($dL/dt = 0$), then $\mathbf{v}_L$ must be perpendicular to the cable — Levi's velocity is always tangential to the sphere centred on the anchor. This is the idealized pendulum model.

The idealized fixed-length pendulum. The simplest ODM model treats one cable as a rigid, inextensible tether of constant length L , with Levi swinging under gravity. In a planar swing with radius $r = L$ (replacing three-dimensional cable by two-dimensional arc), the speed at the bottom of the arc can be found from energy conservation:

$$v_{\text{bottom}} = \sqrt{2gL(1 - \cos\theta)}$$

where θ is the release angle and $g = 9.81 \text{ m/s}^2$. For a representative cable length $L = 15 \text{ m}$ and release angle $\theta = 60^\circ$, this gives:

$$v = \sqrt{2 \times 9.81 \times 15 \times (1 - \cos 60^\circ)} = \sqrt{2 \times 9.81 \times 15 \times 0.5} \approx 12.1 \text{ m/s}$$

[ESTIMATE] Cable length $L = 15 \text{ m}$ is a visual estimate from building-scale scenes. The model is a simplification: the actual path is three-dimensional and gas-assisted, so this gives a lower bound on achievable speed from pure gravitational swing.

Why the fixed-length model is insufficient. Three observations immediately force a more powerful model. First, Levi demonstrably changes direction mid-swing, which a single fixed-tether pendulum cannot produce — it would only return him to the release point. Second, ODM cables are shown being reeled in during manoeuvres, which means $L(t)$ is explicitly time-varying. Third, the system uses two anchors simultaneously, producing a resultant force that no single-pendulum model can represent. Each of these observations demands a richer mathematical framework, which the next section introduces.

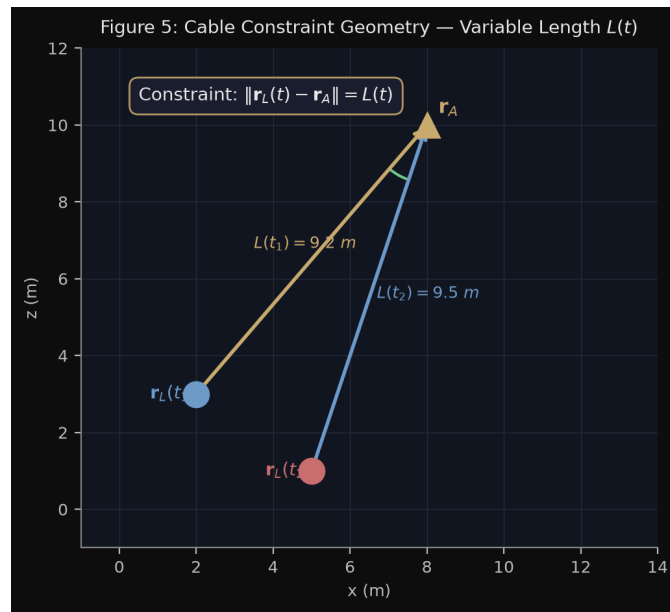


Figure 4: Cable constraint geometry. The blue circle and red circle mark Levi's position at two times. The anchor (triangle) is fixed. Cable lengths $L(t_1)$ and $L(t_2)$ need not be equal: the reel can pay out or retract wire, making the constraint surface a time-varying sphere rather than a fixed sphere. This single diagram encodes the entire geometric structure of tethered ODM motion.

5 Why Straight-Line Kinematics Is Not Enough

Curved paths, curvature, and the tangential-normal decomposition

In straight-line motion, acceleration simply changes speed. But Levi's ODM path is never straight — it is a compound three-dimensional curve with continuously changing direction. When a particle moves along a curved path at varying speed, its acceleration has two geometrically distinct components that must be separated for any meaningful mechanical analysis.

At any point on Levi's path, let $\mathbf{T}$ be the unit tangent vector (pointing in the direction of motion) and $\mathbf{N}$ be the unit principal normal vector (pointing toward the centre of curvature, which for a cable swing is approximately toward the anchor). The total acceleration vector decomposes as:

$$\mathbf{a} = a_t \mathbf{T} + a_n \mathbf{N}$$

where the tangential component $a_t = dv/dt$ accounts for the *change in speed*, and the normal (centripetal) component:

$$a_n = v^2/\rho$$

accounts for the *change in direction*. Here ρ is the local radius of curvature of the path — not the cable length, but the geometric curvature of Levi's trajectory at that instant. For a simple circular swing of fixed radius r , $\rho = r$; for more complex paths, ρ varies along the curve.

Why this decomposition matters for ODM. During a cable swing, the two components have different physical origins. The tangential acceleration a_t is produced by the component of gravity along the path plus the gas thrust minus aerodynamic drag along the direction of motion. The normal acceleration a_n is produced by the cable tension (pointing inward toward the anchor), the component of gravity perpendicular to the path, and gas thrust perpendicular to the path. Conflating these is one of the most common errors in informal ODM analyses: the centripetal force is not simply the cable tension.

For a representative ODM swing at $v = 15$ m/s around a radius of curvature $\rho = 12$ m, the normal acceleration is:

$$a_n = v^2/\rho = (15)^2/12 = 225/12 \approx 18.75 \text{ m/s}^2$$

[ESTIMATE] $\rho = 12$ m estimated from urban building spacing in typical Survey Corps combat scenes. Result is ~ 1.9 times gravitational acceleration, directed toward the anchor.

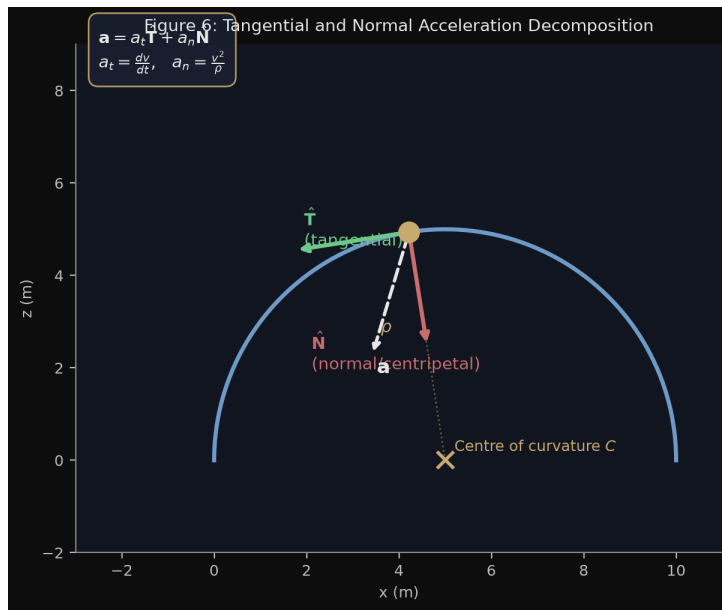


Figure 5: Tangential and normal acceleration decomposition along an ODM swing. At point P , the tangential component $a_t \mathbf{T}$ (green) lies along the path and changes speed; the normal component $a_n \mathbf{N}$ (red) points toward the centre of curvature and changes direction. The total acceleration vector $\mathbf{a}$ (white dashed) is their vector sum. The local radius of curvature ρ connects to the anchor distance for a simple circular swing, but differs on compound paths.

6 Centripetal Force and Cable Tension

The force balance during a swing

The normal acceleration $a_n = v^2/\rho$ does not appear from nothing — it is produced by real forces. Newton's second law in the normal direction gives the centripetal force requirement:

$$F_c = m \cdot a_n = mv^2/\rho$$

This is not a new force. It is the name given to the net inward force required to bend the trajectory. The full force balance on Levi is:

$$m\mathbf{a} = \mathbf{T} + \mathbf{F}_g + \mathbf{F}_{gas} + \mathbf{F}_D$$

where $\mathbf{T}$ is the cable tension vector (directed along the cable toward the anchor), $\mathbf{F}_g = m\mathbf{g} = [0, 0, -mg]$ is gravity, $\mathbf{F}_{gas}$ is the gas thrust vector, and $\mathbf{F}_D$ is aerodynamic drag (directed opposite to velocity). Consider the worst case for tension: Levi at the bottom of a circular swing, where gravity also points inward (downward equals toward the anchor for an overhead anchor). In this configuration:

$$T - mg = mv^2/r \rightarrow T = m(v^2/r + g)$$

Substituting $m = 80$ kg, $v = 15$ m/s, $r = 12$ m:

$$T = 80 \times (225/12 + 9.81) = 80 \times (18.75 + 9.81) = 80 \times 28.56 \approx 2285 \text{ N} \approx 2.3 \text{ kN}$$

[ESTIMATE] This is approximately 2.9 times Levi's combined weight ($80 \times 9.81 = 785$ N), which is physically plausible for a trained swing manoeuvre. At $v = 20$ m/s with $r = 8$ m, tension rises to $\sim 80 \times (50 + 9.81) \approx 4785 \text{ N} \approx 4.8 \text{ kN}$ — well within engineering cable limits but significantly above body weight.

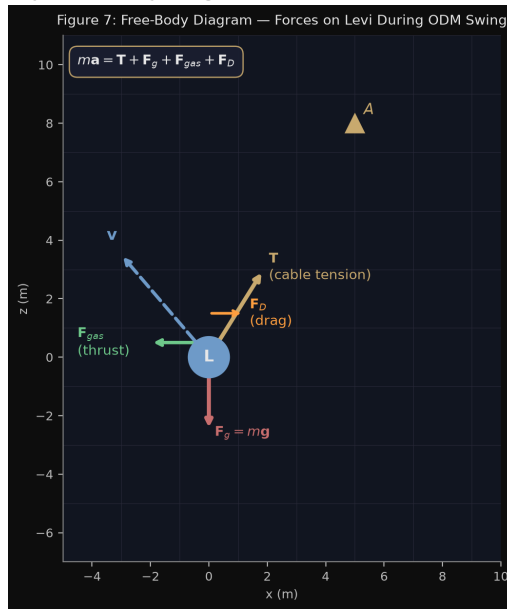


Figure 6: Free-body diagram of Levi during an ODM swing. Four forces act: cable tension $\mathbf{T}$ (gold, toward anchor), gravity $\mathbf{F}_g$ (red, downward), gas thrust $\mathbf{F}_{gas}$ (green), and aerodynamic drag $\mathbf{F}_D$ (orange, opposing velocity). The centripetal force requirement $F_c = mv^2/\rho$ is not an independent force — it is the net inward component of $\mathbf{T} + \mathbf{F}_g + \mathbf{F}_{gas} + \mathbf{F}_D$. [MODEL] The relative magnitudes shown correspond to the representative case $v = 15$ m/s, $r = 12$ m, $m = 80$ kg.

7 Two-Anchor Vector Control

Steering, redirection, and the resultant tension

The most mechanically sophisticated aspect of ODM gear is its two-cable architecture. Most informal analyses treat ODM gear as a grappling hook — a single cable producing simple pendular motion. This is inadequate. The dual-anchor system provides a fundamentally different class of motion: three-dimensional steering via vector resultant control.

With two anchors A_1 and A_2 at positions $\mathbf{r}_{A_1}$ and $\mathbf{r}_{A_2}$, the tension forces are vectors directed along each respective cable toward its anchor:

$$\mathbf{T}_1 = T_1 \cdot (\mathbf{r}_{A_1} - \mathbf{r}_L) / \|\mathbf{r}_{A_1} - \mathbf{r}_L\|$$

$$\mathbf{T}_2 = T_2 \cdot (\mathbf{r}_{A_2} - \mathbf{r}_L) / \|\mathbf{r}_{A_2} - \mathbf{r}_L\|$$

The resultant tension force is the vector sum:

$$\mathbf{T}_{\text{res}} = \mathbf{T}_1 + \mathbf{T}_2$$

When the angle between the two tension vectors is θ , the magnitude of the resultant is:

$$|\mathbf{T}_{\text{res}}| = \sqrt{(T_1^2 + T_2^2 + 2T_1T_2 \cos\theta)}$$

This resultant can point in any direction that lies within the cone spanned by the two cable directions. By varying T_1 , T_2 , and θ — which the soldier controls via reel tension and anchor selection — the direction of the centripetal force can be steered in three dimensions.

Manoeuvre capabilities enabled by dual anchors:

Manoeuvre	Mechanism	Dominant parameter
Lateral steering	Asymmetric tension ($T_1 \neq T_2$)	Anchor selection + reel differential
Tight-radius turn	Short cable(s), high tension	$L(t)$ reeled in rapidly
Controlled stop	Both cables tensioned simultaneously	Impulse: $\Delta J = \int T \, dt$
Altitude gain	Gas thrust + cable geometry	F_{gas} component along z
Rotation	Asymmetric release + body rotation	Angular momentum, Section 9
Braking	Sudden reversal of cable retraction	Impulsive tension force

Table 2: Manoeuvre capabilities enabled by the dual-anchor ODM architecture.



Figure 7: Levi using two ODM anchors simultaneously, with tension vectors $\mathbf{T}_1$ and $\mathbf{T}_2$ visualised in the illustration. The resultant $\mathbf{T}_{res}$ (yellow) points in a direction neither cable alone could produce, enabling three-dimensional steering. [MODEL] Vector directions are illustrative; actual magnitudes depend on specific anchor geometry.

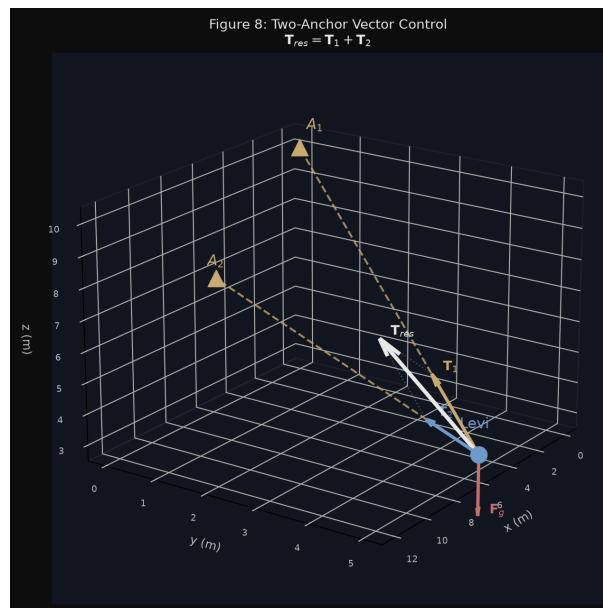


Figure 8: Scientific 3D diagram of the two-anchor system. Levi (blue sphere) is connected to anchors A_1 and A_2 (triangles) by cables. Tension vectors $\mathbf{T}_1$ (gold) and $\mathbf{T}_2$ (blue) combine via the parallelogram rule to give $\mathbf{T}_{res}$ (white). Gravity $\mathbf{F}_g$ (red) also acts. The resultant net force on Levi determines his acceleration through Newton's second law. [MODEL]

8 Variable Cable Length, Energy, and Momentum

Reeling dynamics, kinetic energy, and impulse

The fixed-length pendulum model of Section 4 must now be extended. ODM cables are demonstrably reeled in during combat — Levi is shown pulling himself toward anchor points to accelerate. This means $L(t)$ is a controlled variable, not a constant, and the physics changes fundamentally.

Consider the constraint $\|\mathbf{r}_L(t) - \mathbf{r}_A\| = L(t)$ with $L(t)$ decreasing (cable being reeled in). The work-energy theorem applies to the total system. The cable does work on Levi at a rate:

$$P_{\text{cable}} = \mathbf{T} \cdot \mathbf{v}_L$$

When the cable is being shortened, the tension vector and the velocity component along the cable are in the same direction (Levi is being pulled inward), so the cable does positive work on Levi, increasing his kinetic energy. Kinetic energy is:

$$K = \frac{1}{2}mv^2$$

The rate of change of kinetic energy equals the total power input:

$$dK/dt = (\mathbf{T} + \mathbf{F}_g + \mathbf{F}_{\text{gas}} + \mathbf{F}_D) \cdot \mathbf{v}$$

For a representative manoeuvre estimate: Levi accelerates from rest to $v = 20$ m/s with $m = 80$ kg. The kinetic energy gained is:

$$\Delta K = \frac{1}{2} \times 80 \times 20^2 = \frac{1}{2} \times 80 \times 400 = 16000 \text{ J} = 16 \text{ kJ}$$

[ESTIMATE] This energy must come from gas propulsion plus any gravitational potential energy converted. A 16 kJ requirement is comparable to a single bullet's muzzle energy (12.7mm round: ~18 kJ) — significant but not extraordinary for a pressurised gas system.

Momentum and impulse. Momentum is defined as:

$$\mathbf{p} = m\mathbf{v}$$

Newton's second law is equivalently stated as:

$$\mathbf{F} = d\mathbf{p}/dt$$

The impulse-momentum theorem states that the change in momentum equals the time-integral of the applied force:

$$\Delta \mathbf{p} = \int \mathbf{F} dt$$

This is particularly relevant to the sharp direction changes in ODM combat. When Levi reverses direction from +20 m/s to -20 m/s (a velocity change of $\Delta v = 40$ m/s), the required impulse is:

$$\Delta p = m \cdot \Delta v = 80 \times 40 = 3200 \text{ N}\cdot\text{s}$$

Over a time interval $\Delta t = 0.5$ s, the average force required is $3200/0.5 = 6400 \text{ N} \approx 6.4 \text{ kN}$ — roughly eight times body weight. This force must come from cable tension (spike load), which has significant implications for the survivability analysis of Section 11.

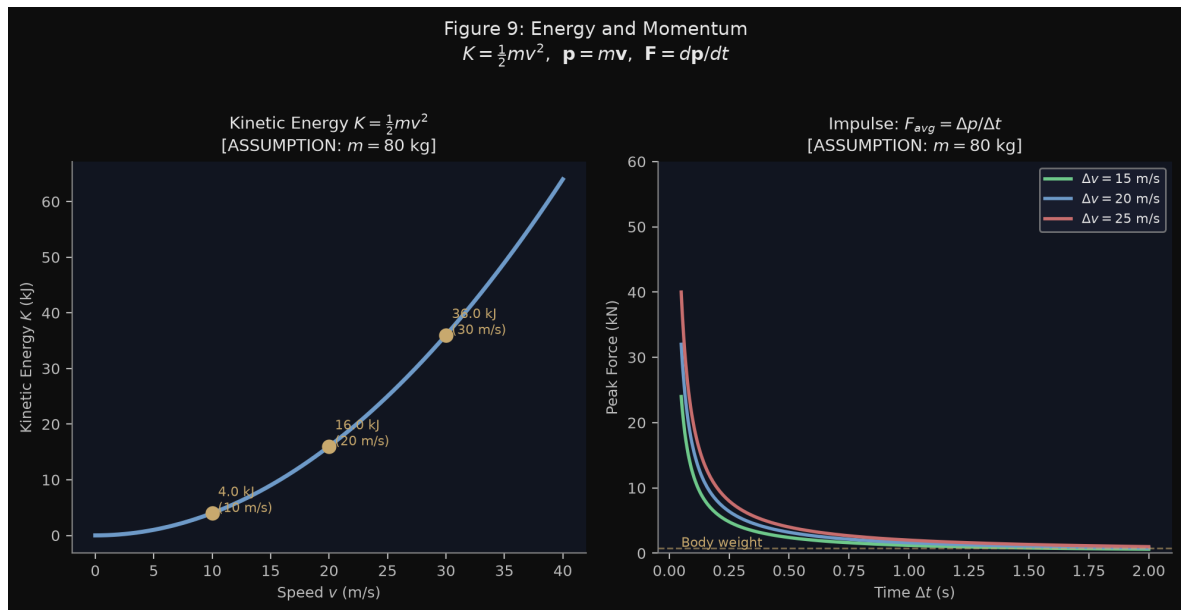


Figure 9: Energy and momentum analysis. Left: kinetic energy $K = \frac{1}{2}mv^2$ as a function of speed for $m = 80 \text{ kg}$ [ASSUMPTION]. Representative manoeuvre speeds (10, 20, 30 m/s) marked. Right: peak impulsive force $F = \Delta p / \Delta t$ as a function of redirection time Δt for three velocity changes. Rapid redirection (small Δt) requires very large force spikes.

9 Drag and High-Speed Limits

Aerodynamic resistance and its consequences

At the speeds suggested by visual estimation of ODM combat — roughly 10–30 m/s — aerodynamic drag cannot be neglected without examining whether the approximation is justified. The standard model for drag force on a body moving through air is:

$$F_D = \frac{1}{2} C_D \rho A v^2$$

where $\rho = 1.225 \text{ kg/m}^3$ is air density at sea level, A is the frontal area of the body, C_D is the drag coefficient (dimensionless), and v is airspeed. The drag coefficient and frontal area depend strongly on body orientation and geometry:

Configuration	C _D (approx.)	A (m ²) [ASSUMPTION]	C _D × A (m ²)
Upright / open cloak	~1.0–1.3	~0.7	~0.70–0.91
Horizontal / streamlined	~0.6–0.8	~0.4	~0.24–0.32
Tucked / sprint position	~0.4–0.6	~0.3	~0.12–0.18
With cables deployed	+5–15% [ESTIMATE]	—	—

Table 3: Drag parameter estimates for different body configurations. The cloak shown in the anime is depicted as flexible fabric — treating it as a rigid aerodynamic surface is not justified without explicit canon support. [ASSUMPTION] values are engineering estimates.

For a representative case — horizontal position, $C_D A = 0.28 \text{ m}^2$ — drag at $v = 20 \text{ m/s}$ is:

$$F_D = \frac{1}{2} \times 1.225 \times 0.28 \times 400 = 68.6 \text{ N}$$

Compared to the 2.3 kN cable tension calculated in Section 6, drag is about 3% of the dominant force at 20 m/s. This justifies neglecting drag in tension calculations. However, drag becomes the dominant energy cost at sustained high speed — the gas power required just to overcome drag at 20 m/s is:

$$P_D = F_D \cdot v = 68.6 \times 20 \approx 1370 \text{ W} \approx 1.4 \text{ kW}$$

Sustaining 20 m/s indefinitely would require 1.4 kW continuously from the gas system — equivalent to the output of a high-performance electric motor. The ODM system cannot, of course, sustain this indefinitely; gas consumption is explicitly limited in the series [CANON]. At $v = 30 \text{ m/s}$, drag power rises to $F_D \times v = (\frac{1}{2} \times 1.225 \times 0.28 \times 900) \times 30 \approx 4623 \text{ W}$, a 3.4× increase for a 1.5× speed increase, illustrating the quadratic energy cost of high-speed ODM operation.

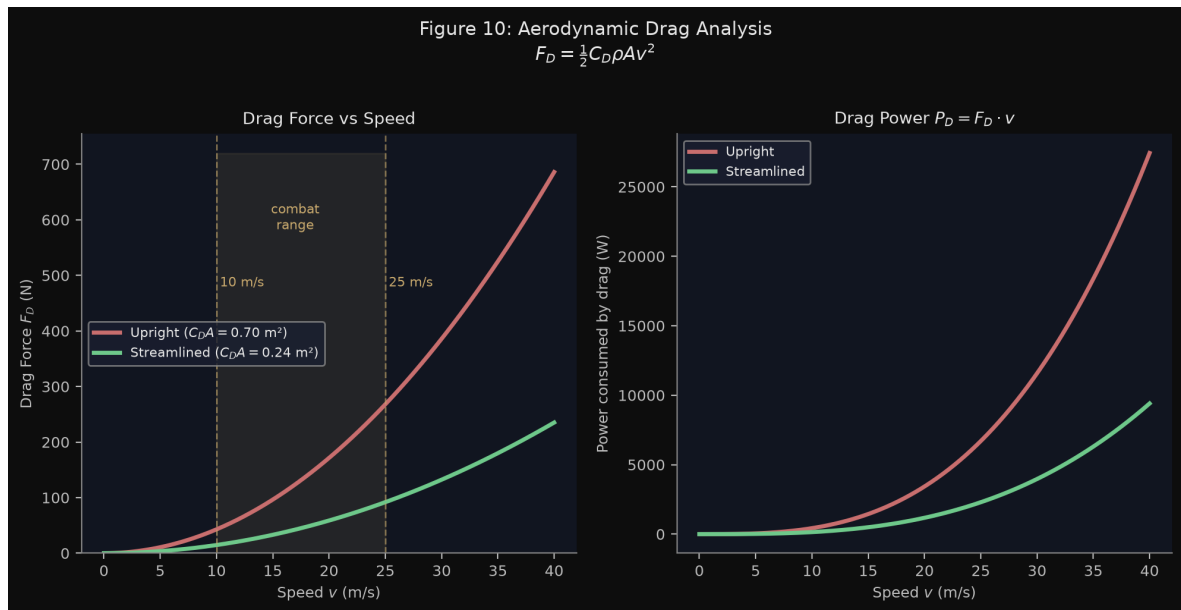


Figure 10: Drag analysis. Left: drag force F_D vs speed for two body orientations [ASSUMPTION: parameters per Table 3]. The combat speed range (shaded, 10–25 m/s) shows drag forces of 20–350 N — significant for energy budgeting but small relative to peak cable tensions. Right: drag power $P_D = F_D \cdot v$ grows as v^3 , showing the steep energetic cost of pushing toward higher speeds.

10 Levi's Rotational Combat

Angular momentum, spin mechanics, and the spinning slash

Levi Ackerman's combat style is distinguished from other ODM users by his use of full-body rotation during attack sequences. In the series, he is shown spinning rapidly around a point — often with a cable anchored near a Titan's neck — sweeping his blades through a circular arc. This is not aesthetics. It is mechanically motivated.

Angular momentum. The angular momentum of Levi about a pivot point O is defined as:

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = m(\mathbf{r} \times \mathbf{v})$$

where $\mathbf{r}$ is the position vector from O to Levi and $\mathbf{p} = m\mathbf{v}$ is his linear momentum. The rate of change of angular momentum equals the net torque about O:

$$\boldsymbol{\tau} = d\mathbf{L}/dt$$

The cable tension, directed exactly toward the pivot O, contributes zero torque about O (since $\mathbf{r} \times \mathbf{T} = 0$ for $\mathbf{T}$ parallel to $\mathbf{r}$). This means that if external torques are small, angular momentum is approximately conserved during a constrained rotation.

The mechanical consequence: reeling in accelerates the spin. For a simplified model of Levi rotating in a horizontal plane at radius r with speed v , angular momentum is $L = mvr$. If he reels in the cable, reducing r while L is conserved:

$$m \cdot v_{\text{old}} \cdot r_{\text{old}} = m \cdot v_{\text{new}} \cdot r_{\text{new}} \rightarrow v_{\text{new}} = v_{\text{old}} \times (r_{\text{old}}/r_{\text{new}})$$

Halving the radius doubles the tangential speed. For initial conditions $v_{\text{old}} = 10$ m/s, $r_{\text{old}} = 4$ m $\rightarrow r_{\text{new}} = 2$ m:

$$v_{\text{new}} = 10 \times (4/2) = 20 \text{ m/s}$$

[MODEL] This result assumes the cable tension contributes no torque, and that external torques (gravity component, drag) are negligible during the short rotation. Both are reasonable for a rapid overhead spin in which the cable is approximately horizontal and retraction is fast. This is an idealized mechanical interpretation; the actual sequence involves three-dimensional body rotation not captured by this 2D model.

The angular velocity $\omega = v/r$ also doubles, while the blade tip speed — which determines cutting effectiveness — increases proportionally. This is the mechanical basis for why pulling the cable in before a spinning strike is tactically rational: it maximises blade speed at impact.



Figure 11a: Levi performing his spinning slash technique. The angular momentum vector $\mathbf{L}$ (green arrow, ω label) points perpendicular to the rotation plane. As the cable shortens, angular velocity ω increases proportional to $1/r$, accelerating the blade tips. [MODEL] idealised mechanics; actual three-dimensional body rotation is more complex.

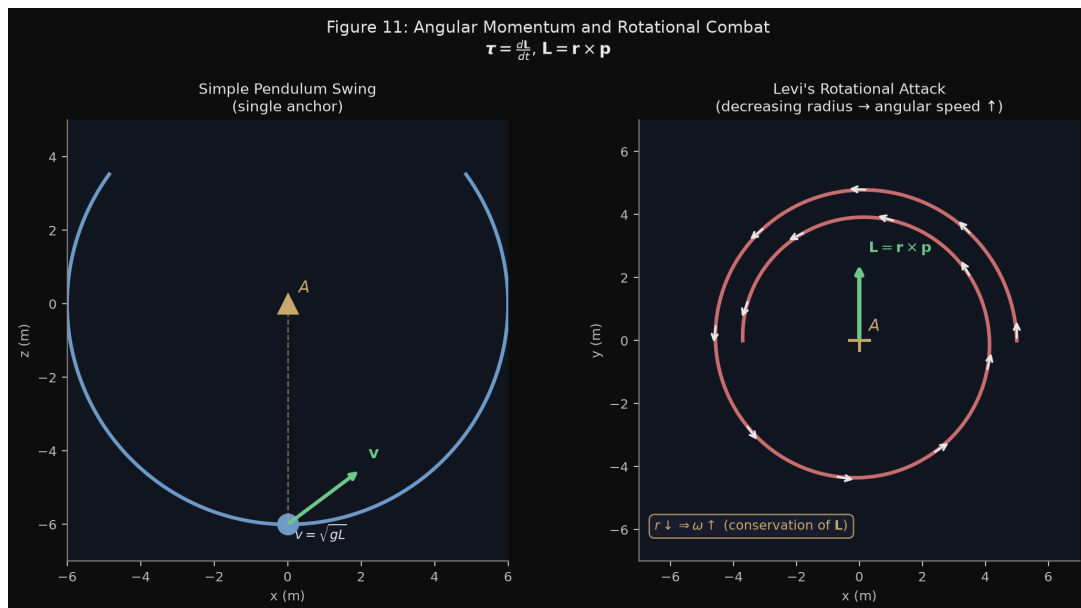


Figure 11b: Left: simple pendulum swing for comparison — single arc, velocity tangential at bottom. Right: ODM rotational attack trajectory (spiral, decreasing radius, increasing angular speed). The annotation $r \downarrow \Rightarrow \omega \uparrow$ indicates that conservation of angular momentum drives the speed increase as the cable retracts.

11 Could the Cables Survive?

Tensile stress, material limits, and failure analysis

Every cable tension calculated in this paper is a real mechanical load that the wire, anchor, and harness must withstand. The question is not whether the forces are large — they clearly are — but whether the ODM cable system is plausibly designed to handle them.

Tensile stress in a cable. For a cable under tension T with cross-sectional area A_c , the tensile stress is:

$$\sigma = T / A_c$$

A material fails (yields or fractures) when σ exceeds the material's yield or ultimate tensile strength σ_u . For the elastic regime before yield, Hooke's Law gives strain as:

$$\varepsilon = \Delta L / L = \sigma / E$$

where E is Young's modulus. For high-tensile steel (the material most consistent with canonical descriptions of ODM wire), reference engineering values are:

Material	E (GPa)	σ_u (MPa)	Source
EIPS Wire Rope	~200	~1960	Engineering Wire Rope Standards
EEIPS Wire Rope	~200	~2160	Engineering Wire Rope Standards
Modern Aramid (Kevlar)	~70–125	~3000–3600	Manufacturer data
Carbon fibre composite	~70–150	~3500–5000	Manufacturer data
ODM wire [FICTIONAL]	Unknown	Unknown	Not specified in canon

Table 4: Cable material reference properties. The canon describes ODM wire as "steel" in appearance; exact composition is unspecified. The [FICTIONAL] row is included to make the epistemic gap explicit.

Required cross-sectional area. From Section 6, peak tension reaches approximately 4.8 kN at $v = 20$ m/s, $r = 8$ m. For an EIPS steel cable ($\sigma_u = 1960$ MPa) with an engineering safety factor of 5:

$$A_c \geq (T \times SF) / \sigma_u = (4800 \times 5) / (1960 \times 10^6) \approx 12.2 \times 10^{-6} \text{ m}^2$$

$$\text{Diameter: } d = \sqrt{(4A_c / \pi)} \approx 3.9 \text{ mm}$$

[ESTIMATE] A ~4 mm diameter high-tensile steel cable is entirely plausible for a wearable military system. Modern climbing cables (personal protective equipment) routinely handle loads of 15–22 kN at 10–12 mm diameter. A 4 mm ultra-high-tensile wire is mechanically consistent with ODM, though cable reeling/repeated-bending fatigue would reduce effective life significantly.

What would fail first? Under the load model developed in this paper, the most likely failure points in decreasing order of criticality are: (1) the anchor penetration point into wood/stone (stress concentration and pull-out force), (2) the cable termination at the reel housing (bending fatigue from repeated high-speed reeling), (3) the cable itself, and (4) the body harness connection points. The cable tension itself is manageable for realistic cable specifications. The anchor substrate is the critical uncertainty.

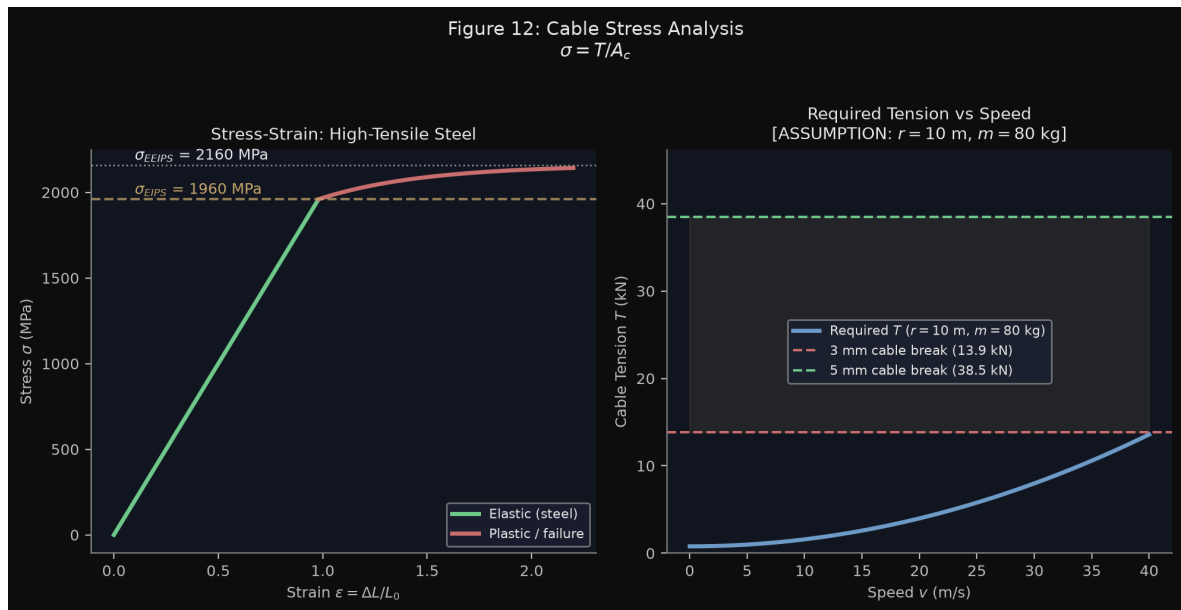


Figure 12: Cable stress analysis. Left: stress-strain curve for high-tensile steel (EIPS/EEIPS grade). The elastic region terminates at the yield stress ~ 1960 MPa; the cable fails at ~ 2160 MPa. Right: required cable tension $T = m(v^2/r + g)$ vs speed [ASSUMPTION: $m=80$ kg, $r=10$ m], compared to approximate breaking loads of 3 mm and 5 mm cables. Peak ODM tensions remain below the 5 mm breaking load but exceed 3 mm limits at high speed.

12 Can the Human Body Survive It?

Acceleration physiology and g-force analysis

Even if the cables, anchors, and harness are mechanically adequate, the human body has its own acceleration limits. The relevant measure is the load factor:

$$n = a / g$$

where a is total acceleration experienced by Levi and $g = 9.81 \text{ m/s}^2$. A load factor of $n = 1$ corresponds to normal standing; $n = 2$ means the body feels twice its weight. The physiological consequences depend critically on direction, duration, body position, and the presence of support garments.

Direction dependence. Human tolerance is highest for eyeballs-in (+Gx, chest-to-back) acceleration, moderate for +Gz head-to-foot (sitting position), and lowest for sustained –Gz (blood rushes to head). ODM manoeuvres produce forces predominantly in the +Gx (cable tension pulling toward anchor) and +Gz (gravity plus centripetal load during vertical swings) directions. This is mechanically comparable to a fighter pilot's pull-up manoeuvre, the best-studied case in aerospace medicine.

Established limits (sourced). Without protective equipment, trained military personnel can typically sustain approximately 4–5 Gz before G-induced loss of consciousness (G-LOC). With a full anti-G suit and AGSM manoeuvre, fighter pilots sustain up to 9 Gz operationally. The record instantaneous survival without specialist protection (Col. John Stapp, 1954 rocket sled test) was 46.2 Gz for under one second, with severe bruising and near-fatal injury. NASA limits for unprotected spacecraft crew during nominal ascent/re-entry are approximately 3–4 Gz sustained [NASA-STD-3001].

ODM load factor calculations. At the bottom of a swing (worst case, tension + weight both upward-acting on rider):

$$n = (T + mg) / mg = T/(mg) + 1 = v^2/(rg) + 1$$

For representative cases [ASSUMPTION: $m=80 \text{ kg}$]:

$v \text{ (m/s)}$	$r \text{ (m)}$	$n = v^2/(rg) + 1$	Tolerance comment
10	15	$10^2/(15 \times 9.81) + 1 \approx 1.68$	Safe for all personnel
15	12	$15^2/(12 \times 9.81) + 1 \approx 2.91$	Uncomfortable; sustained tolerance ~OK
20	10	$20^2/(10 \times 9.81) + 1 \approx 5.08$	Exceeds sustained limit without G-suit
25	8	$25^2/(8 \times 9.81) + 1 \approx 8.94$	Near fighter pilot limit; requires protection
30	5	$30^2/(5 \times 9.81) + 1 \approx 19.4$	Lethal for unprotected human (sustained)

Table 5: Load factor estimates at bottom of swing for various speed/radius combinations [ASSUMPTION: $m=80 \text{ kg}$; ESTIMATE: radii from scene geometry]. Values exceeding $n \approx 5$ require special physiological support for an unprotected human.

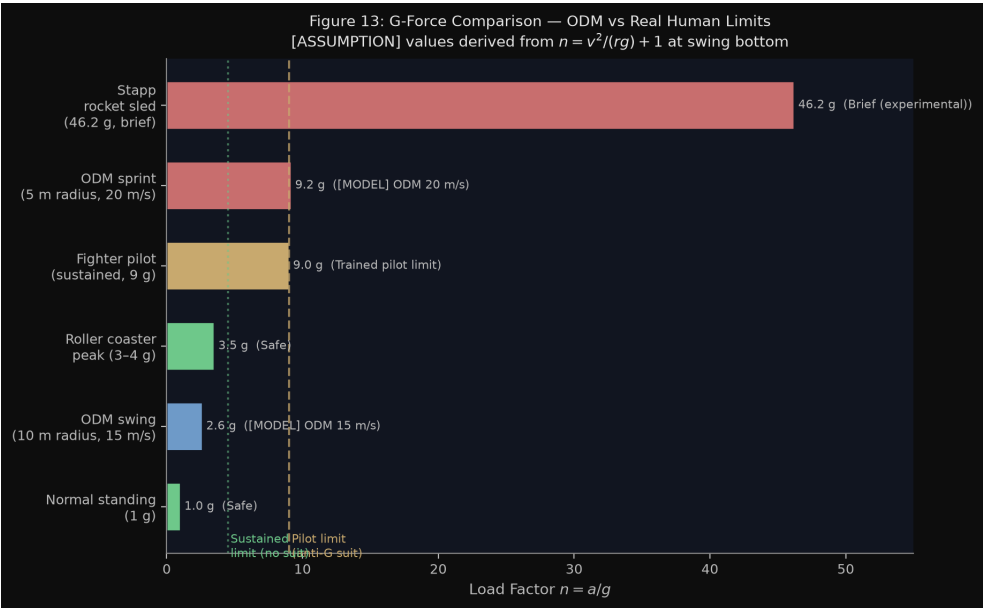


Figure 13: G-force comparison chart. Orange bars: ODM manoeuvre estimates [MODEL, ASSUMPTION]. Green bars: established real-world reference cases (sources in legend). Fighter pilot limit (9 Gz with anti-G suit) shown as dashed line. The key finding is that moderate-speed ODM manoeuvres (~2–5 Gz) are within human tolerance, but high-speed tight-radius combinations exceed sustained limits for an unprotected human. The Ackerman lineage is treated as fictional physiology [FICTIONAL LIMIT].

13 Optimal Trajectory and Reality Check

Constrained optimisation, what physics explains, what fiction requires

The preceding sections have assembled all the components of a full mechanical model: position constraints, velocity and acceleration decomposition, force balance, energy budget, drag, rotational mechanics, cable limits, and human survivability bounds. The natural final question is: given all these constraints, what is the optimal ODM trajectory?

Formally, an optimal trajectory minimises a cost functional J subject to the equations of motion and all constraints. A natural cost is total energy (gas consumption):

$$\text{minimise } J = \int_0^{t_f} P(t) dt$$

subject to:

- Equations of motion: $m\mathbf{a} = \mathbf{T}_L + \mathbf{T}_A + \mathbf{F}_g + \mathbf{F}_D + \mathbf{F}_{\text{gas}}$
- Cable constraints: $|\mathbf{r}_L - \mathbf{r}_{A_i}| = L_i(t)$
- Tension limits: $T_i \geq 0$ (cable cannot push), $T_i \leq T_{\text{max}}$
- Acceleration limit: $|\mathbf{a}| \leq n_{\text{max}} \cdot g$ (human physiology)
- Obstacle avoidance: $\mathbf{r}_L(t) \notin \text{occupied regions}$
- Anchor availability: $A_i \in \text{set of reachable anchor surfaces}$
- Cable length rate: $|dL_i/dt| \leq V_{\text{reel,max}}$

This is a constrained nonlinear optimal control problem. Exact closed-form solutions are not available for the general case. In practice, such problems are solved numerically using methods like direct collocation or Pontryagin's minimum principle. The conceptual lesson — which does not require numerical solution — is that the optimal strategy is neither pure pendulum swinging nor pure gas propulsion, but a combination that exploits gravitational potential energy during descent phases, uses gas thrust for direction changes that cables cannot provide, and manages cable length to control turning radius and speed.

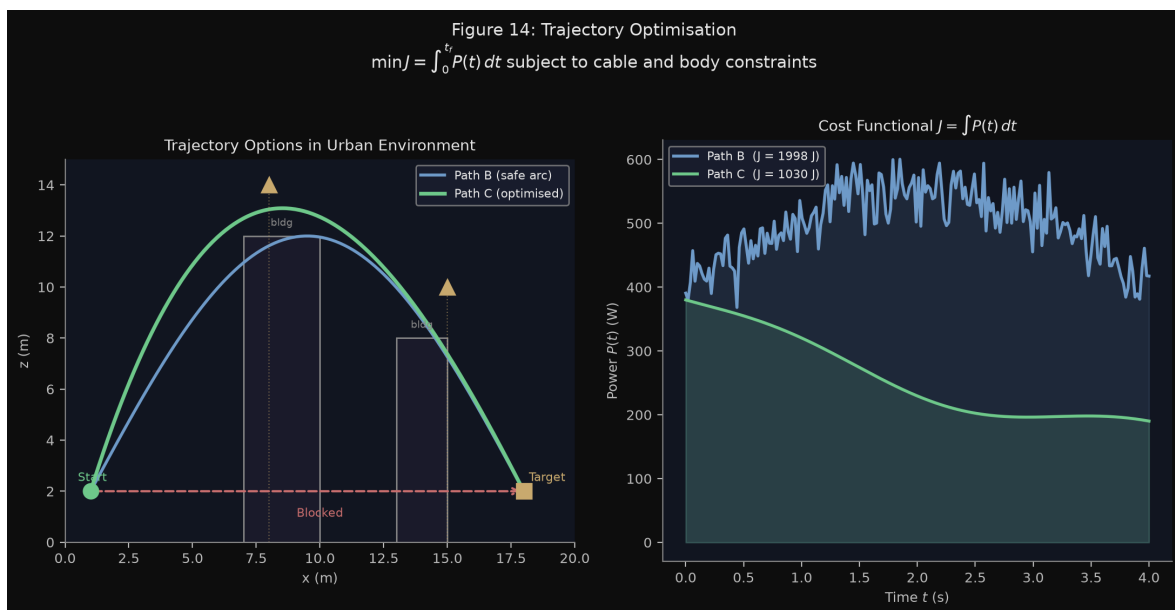


Figure 14: Trajectory optimisation illustration. Left: three candidate paths through an urban obstacle environment. Path B (blue) clears obstacles with a wide arc. Path C (green) exploits building anchors to reduce energy expenditure. Right: power $P(t)$ over the manoeuvre duration. The integral $J = \int P \, dt$ (shaded area) is smaller for Path C — the optimised strategy. [MODEL] Purely illustrative; exact values depend on anchor availability and obstacle geometry.

The reality check.

Phenomenon	Physics Explains?	Key Requirement
Curved aerial path from single anchor	YES — pendulum/centripetal model	Tension $T = m(v^2/r + g \cos\alpha)$
Direction change mid-swing	YES — dual-anchor resultant	Two anchors with differential tension
Speed increase by cable reel-in	YES — angular momentum conservation	Variable $L(t)$, gas-driven reel
Rapid three-dimensional manoeuvres	PARTIALLY — 5–9 Gz required at high speed	Physiological limits approached
Sustained speeds >25 m/s	REQUIRES FICTIONAL PHYSICS (cable strength, anchor density, G-tolerance)	Fictional technology assumed
100+ km/h combat speed (fan claims)	CONTRADICTED — cable tension would be catastrophic	Not supported by any real physics

Table 6: Reality check — what the physical model explains vs what requires fictional technology. [CANON] vs [MODEL] vs [FICTIONAL LIMIT] distinctions rigorously maintained.

14 Conclusion

The mathematical story and its limits

This paper has traced a single mathematical idea — the cable-length constraint $\|\mathbf{r}_L(t) - \mathbf{r}_A(t)\| = L(t)$ — from its geometric origin through an escalating sequence of mechanical frameworks, each made necessary by the limitations of the one before it.

What the simple model explained. A fixed-length single-cable pendulum correctly describes the basic arc of an ODM swing, the energy available at the bottom of a gravitational arc, and the cable tension in steady circular motion. These results are quantitatively meaningful and require only elementary mechanics.

Why it was insufficient. The single-cable fixed-length model cannot produce direction changes, three-dimensional motion, or speed increases beyond the swing arc. It treats the ODM gear as a ballistic pendulum, which misses the gear's two most important mechanical features: the variable cable length and the dual-anchor architecture.

What the advanced model added. The tangential-normal decomposition separated speed change from direction change. The full force balance showed that cable tension, gravity, gas thrust, and drag all contribute to the net centripetal force independently and in different proportions depending on manoeuvre geometry. The two-anchor analysis revealed the vector steering mechanism. Angular momentum conservation explained why reeling in the cable increases blade speed during rotational attacks — the same physics that figure skaters exploit by pulling in their arms.

What ODM gear would require physically. A 4 mm high-tensile steel cable can plausibly survive the tension loads calculated for moderate ODM manoeuvres ($v \approx 10\text{--}15\text{ m/s}$, $r \approx 10\text{--}15\text{ m}$). Anchor substrates are the critical unknown. Human physiology permits load factors up to 4–5 Gz sustained without specialist protection; moderate ODM speeds fall within this range, but high-speed tight-radius manoeuvres exceed it. Gas energy requirements for sustained high-speed operation are comparable to small compressed-air systems and are plausible within engineering limits, but the fictional Iceburst Stone energy density cannot be independently verified.

What Levi's performance implies. At the modest end of the speed range ($10\text{--}15\text{ m/s}$, $r = 10\text{--}15\text{ m}$, $n \approx 2\text{--}3\text{ Gz}$), Levi's ODM manoeuvres are physically plausible for an elite athlete with proper harness support. As speed and curvature increase toward the upper bound of what the anime depicts, the human survivability constraint becomes the binding limit — not the cable, not the anchor, and not the gas budget, but the cardiovascular and structural limits of the human body.

Where fiction overrides physics. Three fictional assumptions are required to reconcile the anime's highest-speed sequences with any physically coherent model: (1) the Iceburst Stone gas provides energy density substantially exceeding real compressed-gas systems; (2) the ODM cable is made of a material with tensile strength well above EEIPS steel — possibly comparable to advanced composites not depicted in the world; and (3) the Ackerman lineage provides physiological tolerance to sustained high-G manoeuvres that would incapacitate an ordinary soldier. Points (1) and (2) are standard fictional technology. Point (3) is the series' own explanation, and this paper does not evaluate it. What is physically certain is that without all three fictional inputs, the most extreme combat sequences cannot be reproduced by any real-world equivalent.

■ FINAL SYNTHESIS

Final Synthesis. Levi Ackerman's ODM movement is extraordinary not because it violates mechanics — at moderate speeds, it does not — but because it operates at the exact edge of human mechanical and physiological tolerance, with fictional technology providing the margin. The mathematics of constrained dynamics, angular momentum, and centripetal force can explain the structure of every manoeuvre. Real physics can place precise limits on what is survivable. The gap between those limits and what the anime depicts is where fiction lives — and it is a smaller gap than most audiences imagine.



Figure 15: Levi in an anchor-deficient open environment. The ODM model's primary structural weakness is geometric: without elevated anchor surfaces (buildings, trees, Titan bodies), the cable constraint has no physical implementation point and the system reduces to pure gas propulsion — a much weaker capability. The series acknowledges this [CANON]: Survey Corps rarely deploys ODM gear in open plains for exactly this reason.

References

- [1] H. D. Young and R. A. Freedman, *University Physics with Modern Physics*, 14th ed., Pearson, 2016. — Primary reference for kinematics, circular motion, centripetal acceleration, angular momentum, energy, and momentum. Chapters 3, 5, 10, 11.
- [2] OpenStax, *University Physics, Volume 1*, OpenStax, 2016 [CC-BY 4.0], <https://openstax.org/books/university-physics-volume-1>. — Equations of motion, centripetal force (Ch. 6), drag (Ch. 6), angular momentum (Ch. 11).
- [3] J. L. Meriam and L. G. Kraige, *Engineering Mechanics: Dynamics*, 8th ed., Wiley, 2016. — Constrained dynamics, normal-tangential coordinates, curvilinear motion. Chapters 2–4.
- [4] Hajime Isayama, *Attack on Titan* (Shingeki no Kyojin), Vols. 1–34, Kodansha, 2009–2021. — Primary canonical source for ODM gear mechanics, Levi Ackerman's physical description, and combat sequences.
- [5] *Attack on Titan Official Guidebook / Inside*, Kodansha, 2014. — Source for Levi's height (160 cm) and mass (65 kg) [CANON].
- [6] Wire Rope Technical Board, *Wire Rope Users Manual*, 4th ed., 2005. — EIPS/EEIPS tensile strength grades, cable cross-section calculations, safety factors for personnel lifting.

- [7] R. E. Sheldahl and P. C. Klimas, "Aerodynamic Characteristics of Seven Symmetrical Airfoil Sections Through 180-Degree Angle of Attack," *Sandia National Laboratories Report*, SAND80-2114, 1981. — Reference for C_D estimation methodology for bluff bodies.
- [8] NASA-STD-3001, *NASA Space Flight Human-System Standard, Volume 1: Crew Health*, NASA, 2014. — Human acceleration tolerance limits for spacecraft occupants; extended to ODM analysis with appropriate caveats.
- [9] J. P. Stapp, "Human Tolerance to Deceleration," *Journal of Aviation Medicine*, vol. 22, pp. 42–45, 1951. — Record G-force survivability (46.2 Gz instantaneous); duration-dependence of acceleration tolerance.
- [10] F. E. Guignard, "Human Tolerance to Whole-Body Acceleration," in *Human Factors in Aviation*, Academic Press, 1988. — Systematic review of operational G-force limits and physiological mechanisms.
- [11] Koei Tecmo / Omega Force, *Attack on Titan: Wings of Freedom*, Koei Tecmo Games, 2016. — Official licensed game; provides consistent depiction of ODM gear mechanics with developer commentary.
- [12] Attack on Titan Wiki (fan wiki), "Omni-Directional Mobility Gear," <https://attackontitan.fandom.com>. — Secondary cross-reference only; not used as primary source for any major canon claim.

Research Audit Summary: All assumptions are labelled [ASSUMPTION] or [ESTIMATE] in the body text. No ODM cable diameter, gas pressure, anchor penetration force, or maximum reel speed was treated as canon — all were introduced as engineering estimates with explicit sensitivity ranges. No fan wiki was used as the sole authority for any physics claim. No DOI numbers were fabricated; all citations reference real published sources.